

Answers to Questions 1-16

Q1:

1. Function of a surge protector in electrical installations:

A surge protector protects electrical machines from voltage spikes or surges by diverting excess voltage safely to the ground, thus preventing damage.

Q2:

2. Purpose of grounding electrical machines:

Grounding provides a safe path for fault currents to flow to the earth, reducing the risk of electric shock and protecting equipment from damage.

Q3:

3. High current in electrical circuits and results in fire (True or False):

True: High currents can cause overheating of wires and insulation, potentially leading to fire hazards.

Q4:

4. Machine alignment definition:

Machine alignment refers to positioning the motor and the load in a straight line to ensure smooth operation and minimize wear and tear on components.

Motor couplings should be aligned (True or False): True. Misaligned couplings lead to excessive vibration, wear, and potential machine failure.

Q5:

5. Minimum insulation resistance test value:

For electrical installations, the minimum insulation resistance test result should not be less than 1 mega-ohm.

Q6:

6. Definitions regarding electrical installation:

- Inspection: Checking electrical systems or installations to ensure compliance with safety standards.

- Testing: Performing tests on electrical systems to verify proper operation and safety.

Q7:

7. Role of a thermal overload relay:

A thermal overload relay protects electric motors by interrupting the circuit when excessive heat or current is detected.

Q8:

8. Difference between an electric motor and a generator:

- Electric Motor: Converts electrical energy into mechanical energy.
- Electric Generator: Converts mechanical energy into electrical energy.

Q9:

9. Four factors considered in machine layout design:

- Space availability.
- Ease of maintenance.
- Cooling and ventilation.
- Noise and vibration control.

Q10:

10. Three advantages of transformers over electrical machines:

- High efficiency.
- Low maintenance requirements.
- Ability to transfer power over long distances.

Q11:

11. Four factors affecting motor efficiency:

- Quality of materials used.
- Design of the motor.

- Operating conditions (e.g., load).
- Maintenance and lubrication.

Q12:

12. Functions of components in a DC machine:

- Brush: Conducts current between the stationary and rotating parts of the machine.
- Commutator: Converts AC generated in the armature winding to DC for external use.

Q13:

13. Three tools used in cable termination:

- Cable crimper.
- Wire stripper.
- Heat shrink tubing or insulation tape.

Q14:

14. Three tests performed after motor installation:

- Insulation resistance test: Verifies the insulation quality.
- No-load test: Ensures proper operation without any load.
- Alignment test: Confirms correct motor alignment with the load.

Q15:

15. Four factors considered when selecting an electric motor for industrial drives:

- Load requirements (e.g., torque and speed).
- Type of application (e.g., continuous or intermittent operation).
- Environmental conditions (e.g., temperature and humidity).
- Efficiency and power consumption.